

Problem F. Venn Diagram

Time Limit 20000 ms

Mem Limit 524288 kB

Problem Statement

Alice is a private teacher. One of her job is to prepare the learning materials for her student. Now, as part of the materials, she is drawing a Venn diagram between two sets A and B .

Venn diagram is a diagram which illustrates the relationships among one or more sets. For example, a Venn diagram between two sets A and B is drawn as illustrated below. The rectangle corresponds to the universal set U . The two circles in the rectangle correspond to two sets A and B , respectively. The intersection of the two circles corresponds to the intersection of the two sets, i.e. $A \cap B$.

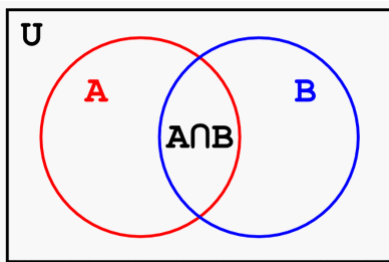


Fig: Venn diagram between two sets

Alice, the mathematics personified, came up with a special condition to make her Venn diagram more beautiful. Her condition is that the area of each part of her Venn diagram is equal to the number of elements in its corresponding set. In other words, one circle must have the area equal to $|A|$, the other circle must have the area equal to $|B|$, and their intersection must have the area equal to $|A \cap B|$. Here, $|X|$ denotes the number of elements in a set X .

Alice already drew a rectangle, but has been having a trouble figuring out where to draw the rest, two circles, because she cannot even stand with a small error human would make. As an old friend of Alice's, your task is to help her by writing a program to determine the centers and radii of two circles so that they satisfy the above condition.

Input

The input is a sequence of datasets. The number of datasets is not more than 300.

Each dataset is formatted as follows.

$$U_W \ U_H \ |A| \ |B| \ |A \cap B|$$

The first two integers U_W and U_H ($1 \leq U_W, U_H \leq 100$) denote the width and height of the rectangle which corresponds to the universal set U , respectively. The next three integers $|A|$, $|B|$ and $|A \cap B|$ ($1 \leq |A|, |B| \leq 10,000$ and $0 \leq |A \cap B| \leq \min\{|A|, |B|\}$) denote the numbers of elements of the set A , B and $A \cap B$, respectively. The input is terminated by five zeroes.

You may assume that, even if U_W and U_H would vary within ± 0.01 , it would not change whether you can draw two circles under the Alice's condition.

Output

For each dataset, output the centers and radii of the two circles that satisfy the Alice's condition as follows:

$$X_A \ Y_A \ R_A \ X_B \ Y_B \ R_B$$

X_A and Y_A are the coordinates of the center of the circle which corresponds to the set A . R_A is the radius of the circle which corresponds to the set A . X_B , Y_B and R_B are the values for the set B . These values must be separated by a space.

If it is impossible to satisfy the condition, output:

impossible

The area of each part must not have an absolute error greater than 0.0001. Also, the two circles must stay inside the rectangle with a margin of 0.0001 for error, or more precisely, all of the following four conditions must be satisfied:

- $X_A - R_A \geq -0.0001$
- $X_A + R_A \leq U_W + 0.0001$
- $Y_A - R_A \geq -0.0001$
- $Y_A + R_A \leq U_H + 0.0001$

The same conditions must hold for X_B , Y_B and R_B .

Sample Input

```
10 5 1 1 0
10 5 2 2 1
10 10 70 70 20
0 0 0 0 0
```

Output for the Sample Input

```
1 1 0.564189584 3 1 0.564189584
1 1 0.797884561 1.644647246 1 0.797884561
impossible
```